Paris, France
(+33)6 - 50 - 84 - 94 - 62
sebastiengradit@gmail.com

Sébastien **Gradit**

Scientific Consultant AI Biotech

sebgra
sebastiengradit.com
sebastien-gradit

Computational Biologist with 5+ years of experience in multi-omics integration and high-fidelity ML frameworks. I specialize in transforming complex genomic data into actionable biological narratives to accelerate drug discovery and precision oncology. By building reproducible, industrial-grade R&D platforms, I bridge the gap between raw multi-modal data and the clinical insights needed for pharma-partnered projects.

Skills

Data Science	**Machine Learning** [*Keras*, *TensorFlow*, *PyTorch*, *SciKit-Learn*], **SQL**
Bioinformatics	**Multi-Omics Integration**[*Genomics, Proteomics*], **High Throughput Sequencing**, **Clinical Imaging**
Core Programming	**Python**, **C++**, **R**, **High Performance Computing** [*Slurm*], **Cloud** [*AWS*]
Data Management	**CI/CD** [*GitHub Actions, Snakemake*], **Containerization** [*Docker, Singularity*], **Versioning** [*Git, GitHub*]

Experience

UVSQ – Université de Versailles Saint-Quentin-en-Yvelines (IHU SEPSIS) *Versailles, France*

Data Scientist – Metabolomics & Machine Learning *2026*

- Engineered robust, automated signal processing and batch-correction pipelines for large-scale LC-MS metabolomics datasets comprising over 1,000 clinical samples.
- Implemented rigorous statistical learning and multivariate analysis frameworks to tackle high-dimensional (5000+ metabolic features) biological data for sepsis diagnostic signature discovery.
- Refactored and optimized exploratory legacy codebases into modular, FAIR-compliant workflows, reducing processing time by 15% and ensuring high reproducibility across clinical research teams.

Independent R&D *Paris, France*

Machine Learning & Bioinformatics Researcher *2025*

- Developed SynBioGuard, a protein safety classifier utilizing biological foundation models (LLMs) to assess synthetic designs. Focused on model interpretability to translate sequences into biosecurity risk assessments..

Pasteur Institute *Paris, France*

Bioinformatician Researcher | Bioinformatics and machine learning *2021 – 2025*

- Pioneered ”Hicberg” an ML-driven Python framework that recovers 100% of generated sequencing data, achieving a 30% information gain in repetitive genomic regions compared to existing methods.
- Engineered a platform-agnostic architecture with 95% validated fidelity against ground-truth, ensuring high-fidelity reconstruction across genomic data types.
- Unlocked previously ”unexploitable” therapeutic targets, providing a first-of-its-kind roadmap for precision medicine R&D.
- Architected industrial-grade R&D infrastructure (Snakemake, Docker, Slurm) to automate high-throughput processing on HPC clusters with 100% reproducibility.

Siemens HealthCare - LIPADE *Paris, France*

Biomedical Engineer *2020*

- Engineered a high-performance C++ feature extraction engine that automated bone lesion classification in xSPECT imaging, accelerating diagnostic workflows by 25% and eliminating manual contouring subjectivity for radiologists.
- Engineered an embedded ML pipeline for clinical volume segmentation, focusing on explainable AI features to support radiologist decision-making and eliminate manual contouring subjectivity.

Education

PhD in Bioinformatics, Pasteur Institute *Paris, France* *2021 - 2025*

M.Sc. in Image Processing and Artificial Intelligence - GPA: 3.33/4.0, Télécom Paris - Sorbonne University *Paris, France* *2019 - 2020*

M.Sc. in Biomedical Engineering (Engineer Diploma.) - GPA: 4.0/4.0, EPISEN ISBS - ESIEE *Paris, France* *2016 - 2019*

Soft Skills

Soft Skills

- Analytical Thinking & Complex Problem-Solving
- Adaptability, Agility & Continuous Learning
- Creative Thinking & Innovation
- Effective Communication & Multidisciplinary Collaboration

Honors & Awards

2025 **1st Place**, Read Write Grow - Biocatalyst Foundation *Europe, Remote*
Classification of protein sequence safety through LLM and AI for Synthetic Biology designs

2024 **INCEPTION Program Grant**, Fellowship *Paris, France*
Awarded a highly competitive grant covering the final three months of the PhD program, demonstrating recognition of research merit and strategic value to the field.

2022 **1st Place**, Digital 4 Genomics Hackathon *Genopole, Evry*

Certificates

2023 **TensorFlow Advanced Specialization**, DeepLearning.AI

2023 **Deep Learning Specialization**, DeepLearning.AI

2023 **Machine Learning Specialization**, Stanford - DeepLearning.AI

Publications

Hicberg: Reconstruction of Contact Signals from Repeated Elements, **Genome Research** - *2026*
Gradit S., *Ortion S.*, *Larrous P.*, *Delouis M.*, *Koszul R.*, *Cournac A.* - doi: 10.1101/gr.281770.125

Selected scientific communications

2024 **JOBIM 2024, Toulouse**, Poster | Prediction of Omics Signal from Repeated Elements

2024 **JeBiF@JOBIM 2024, Toulouse**, Workshop | Good Practices in Bioinformatics

2023 **International Congress for Transposable Elements, St-Malo**, Poster | Prediction of Omics Signal from Repeated Elements

2023 **31st Intelligence Systems for Molecular Biology (ISMB), Lyon**, Poster and Talk | Statistical Inference of Repeated Elements Contacts in Hi-C maps

Extracurricular Activity

Jeunes BioInformaticiens de France (JeBiF) *France*
President *2022 - 2024*

- Led strategic initiatives to **structure and energize** France's emerging bioinformatics talent.
- Fostered **national and international collaborations** within the bioinformatics community.
- **Promoted and popularized** bioinformatics to public, private, and international stakeholders.
- Organized **community-building events** and disseminated information on bioinformatics training.